

Interface Capability Brief

EDI 278 Prior Auth Demo · Stakeholder overview · 2026-08-07

1. Executive summary

Demo of opportunity idea 2.1 Prior authorization that does not die in fax and portal hell: ingest X12 278 prior-auth requests, apply completeness rules (member, procedure, diagnosis, attachment when required), simulate a payer decision, and emit an X12 278 response plus an EHR HL7 ORU “auth decided” notice. Demo only — not Da Vinci PAS, not real payer APIs.

2. Who this is for

- Sources: Provider PA drop folder (input/inbound/.edi) - Destinations: decision buckets (approved / denied / incomplete / pending), output/responses/.edi (278), output/ehr-notices/.hl7 (ORU), SQL AuthRequest status - Demo vs production: Demo only — synthetic 278, catalog rules in SQL, no FHIR PAS path in v1, no LLP to a live EHR

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3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

- Transport: Directory / File (DirectoryListener)
- Format / envelope: X12 278 (5010-shaped Health Care Services Review Request), multi-ST / multi-Transaction
- Identity fields: TRN02 → AuthTraceNumber (also keyed on demo catalog outcomes)
- Samples path: samples/

Integration highlights

FQCN sources: Sandbox demos (edi-835-payment-integrity, medical-device-hl7-ehr) + playbook. Image: pilotfish-eip:23R1.

Stage	Module / mechanism	Notes
R1 Listener	DirectoryListener	Poll \$\$EDI_INBOUND_DIRECTORY
R1 Format	EDITransformationModule + XPathForkingModule //Transaction	Split each ST; UseInternalData=false (23R1 trial X12 tables expired)
R1 Target processors	XPath extract → DatabaseSqlProcessor AuthCatalog lookup → XSLT completeness + disposition	Build AuthDecision
R1 Transport	DirectoryTransport	Stage under output/staged-decisions/
R2 Listener	DirectoryListener	Poll staged decisions
R2 Router	XPathRoutingModule accumulate=true	Emit 278+ORU+SQL for all; bucket writers by DecisionBucket
R2 Transports	DirectoryTransport + NullTransport + SQL UPDATE	Files + AuthRequest status

4. Honest boundaries (not claimed yet)

- Add FHIR Da Vinci PAS inbound path in a later pass?
- LLP delivery of ORU instead of file drop?

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

- Exercise route “1 - Intake And Completeness” from the Web UI or documented smoke path
- Exercise route “2 - Decide And Emit Responses” from the Web UI or documented smoke path
- Confirm outputs land in the documented folders / queues and appear in the Web UI status views

6. How it works (plain language)

7. Validation

- What is checked: EDI parses; TRN identity present; completeness (MemberId, PatientName, ProcedureCode, DiagnosisCode; attachment when catalog requires); catalog disposition for complete requests - What is NOT checked: full 278 SNIP, HCD/UM loop fidelity, real clinical attachment content, Da Vinci PAS, LOINC questionnaires - Does failure block outbound? yes for unparseable EDI; no for incompleteness (incomplete bucket + 278/ORU still emit)

6. State & idempotency

- Status model: AuthRequest.Status OPEN → APPROVED | DENIED | INCOMPLETE | PENDED - When state advances: after AuthDecision is routed - Dedup keys: AuthTraceNumber + inbound file stamp (resubmits overwrite decision / response files) - Retry / poison: bad EDI fails at EDI→XML (fail-closed); incompleteness is business theater (does not hard-fail the route)

8. Dual-write / side effects

- Order: stage AuthDecision → route fan-out (278 + ORU + bucket + SQL) - Compensation: none (demo) - Demo shortcuts: simulated payer is SQL DefaultDisposition + completeness gate; attachment is a PWK / REF*EA flag, not a real 275 payload

3. Inbound contract

- Transport: Directory / File (DirectoryListener) - Format / envelope: X12 278 (5010-shaped Health Care Services Review Request), multi-ST / multi-Transaction - Identity fields: TRN02 → AuthTraceNumber (also keyed on demo catalog outcomes) - Samples path: samples/

Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

1 - Intake And Completeness: runs steps such as Poll EDI 278 Files, Fork - XPath, Extract 278 Auth Fields, Debug Write Forked Transaction.

2 - Decide And Emit Responses: runs steps such as Poll Staged Auth Decisions, Extract Decision Attributes, Conditional Router, Map Decision To 278 Response.

7. Security & trust posture

- X12 SNIP validation is used where configured to reject bad EDI before downstream work
- Credentials and passwords in this package are demo-only — not production secrets
- Risk noted in design: High — No real payer / FHIR PAS — Pitch lists Da Vinci + portals

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- Risk noted in design: Med — Trial X12 tables expired on 23R1 — UseInternalData=false; generic Segment/Element XML
 - Risk noted in design: Med — Claim-before-complete on multi-file emit — 278, ORU, bucket, SQL not XA
 - Risk noted in design: Med — Synthetic 278 not clearinghouse-certified — Theater segments only
 - Risk noted in design: Low — Multi-service loop per ST — Fork is //Transaction

8. Seeing it work

- - Ports: SQL 14340, EIP 8120, Web UI 8121 - Volumes: ./input, ./output, ./logs, ./samples, ./documents - Heap: 1–2GB - Dependencies / cold start: SQL health + seed ~30–60s, EIP ~60–90s - LAN: Web UI on 0.0.0.0; LAN_HINT=http://192.168.68.52:8121/
- docker compose up -d --build
- - Web UI (local): http://localhost:8121/
- - PilotFish EIP: http://localhost:8120/eip/
- - Route diagrams: http://localhost:8121/documents/route-diagrams.pdf ·
http://192.168.68.52:8121/documents/route-diagrams.pdf
- - Capability brief: http://localhost:8121/documents/capability-brief.pdf ·
http://192.168.68.52:8121/documents/capability-brief.pdf
- docker compose logs -f pilotfish
- docker compose exec sqlserver /opt/mssql-tools18/bin/sqlcmd \
- -S localhost -U sa -P 'PilotFish_Demo1!' -C -d Edi278PriorAuth \
- docker compose down -v

9. Related technical documents

- DESIGN.md — working engineering specification for this interface
- README.md — how to run, ports, and smoke commands
- This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.